Josephson effect

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1 Introduction

Superconductivity produces a variety of intriguing results. In this experiment, the interaction of superconductors is investigated. If two superconductors are loosely linked, the Josephson effect occurs. The Josephson effect is an example of a quantum effect at the macroscopic level. Josephson contacts are utilized for quick electrical switching and measuring B-fields to the accuracy of individual flux quantums.

2 Preparation

2.1 Critical Temperature

Some materials become superconducting at temperatures below their critical temperature (T_c). During this condition, the material's properties shift significantly. The most well-known example is the electric resistance, which rapidly decreases to insignificant levels. This allows for almost infinitely lasting current loops in the superconductor. Its magnetic properties change drastically as well acting as a perfect diamagnet expelling all magnetic fields [6].

2.2 Cooper pair

It involves so-called Cooper pairs. As an electron travels through the atomic lattice, it displaces the atomic cores from their equilibrium positions, creating a region of net positive charge along its path. This movement induces a dynamic polarization in the lattice. A second electron is then drawn to the positive charge concentration caused by the first electron, and vice versa, leading to a mutual attraction between the two electrons mediated by the lattice polarization. This attractive interaction allows an electron to reduce its energy by moving through an already polarized lattice. The energy reduction is greatest when the two electrons travel in opposite directions with equal momentum and opposite spins, forming a Cooper pair [2, Chapter 10.3]. Their interaction is facilitated by phonons, the quantized vibrational modes of the lattice. Typical coherence lengths for Cooper pairs are on the order of $\xi = 3000\,\text{Å}$ [6, p. 11].

2.3 Meissner–Ochsenfeld effect

Superconductors exhibit unusual behavior in magnetic fields compared to normal conductors, like for example metals. When a perfect conductor, which has zero electrical resistance ($R = 0$) below a critical temperature ($T < T_c$), is exposed to an external magnetic field, a unique phenomenon is observed for a closed current loop encircling an area A.

$$I \cdot R = U = \int_A \nabla \times \vec{E} \cdot d\vec{A} = -\frac{d\vec{B}}{dt} \cdot \vec{A}$$

When $R = 0$, it implies that the dot product $\vec{B} \cdot \vec{A}$ must remain constant, leading to two distinct scenarios:

First Scenario (Cooldown after external magnetic field activation): The conductor is initially

placed in a magnetic field, cooled below T_c, and then the magnetic field turned off. This process generates currents within the conductor that sustain the magnetic field described by Lenz's law.
Second Scenario (Cooldown before external magnetic field activation): The conductor is positioned in an environment free of magnetic fields, cooled below T_c, and then a magnetic field activated. This triggers currents within the conductor that oppose the external magnetic field, effectively canceling it inside the conductor. When the external field is deactivated, the conductor returns to a field-free state.

Thus, a perfect conductor can exist in two distinct states depending on its history. However, this behavior is not seen in superconductors. Superconductors consistently maintain a field-free interior at temperatures below T_c, regardless of their prior magnetic state at higher temperatures. Therefore inside a superconductor $\vec{B} = 0$, a phenomenon known as the Meissner-Ochsenfeld effect. Consequently, superconductors act as perfect diamagnets. This behavior was first explained by the London brothers in 1935, who provided the equation of motion for superconducting electrons as follows by assuming the phenomenological τ-relaxation time going to infinity from the standard Drude model of conductors ($m\frac{d\vec{v}}{dt} = -e\vec{E} - m\vec{v}/\tau$)[6] :

$$m\frac{d\vec{v}}{dt} = -e\vec{E}$$

Given the supercurrent density $\vec{j}_s = -en_s\vec{v}$, where n_s is the electron density, the first London equation is derived as follows:

$$\frac{\partial \vec{j}_s}{\partial t} = \frac{\vec{E}}{\lambda_L} \quad \text{with} \quad \lambda_L = \frac{m}{n_s e^2}$$

When combined with the Maxwell equation $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$, it is deduced that:

$$\frac{\partial}{\partial t}\left(\lambda_L \nabla \times \vec{j}_s + \vec{B}\right) = 0 \quad \Rightarrow \quad \lambda_L \nabla \times \vec{j}_s + \vec{B} = \text{const.}$$

This equation accounts for the Meissner-Ochsenfeld effect only when the constant is zero, leading to the second London equation [3, Chapter 2-3]:

$$\nabla \times \lambda_L \vec{j}_s = -\vec{B}$$

Using the Maxwell equation $\nabla \times \vec{B} = \mu_0 \vec{j}_s$, this leads to a differential equation, using $\nabla \cdot B = 0$:

$$\Delta \vec{B} - \frac{\mu_0}{\lambda_L}\vec{B} = 0$$

with the solution

$$B_x(z) = B_x(0)e^{-z\sqrt{\mu_0/\lambda_L}}$$

In the 2-dimensional scenario where $\vec{B} = B_\mathrm{x}\hat{\mathrm{e}}_\mathrm{x}$, the London penetration depth $\Lambda_L = \sqrt{\mu_0/\lambda_L}$ is defined as the distance at which the B-field decreases to $1/e$ of the B-field at the surface. This means that the magnetic field decreases exponentially as one gets deeper into the superconductor. It can be interpreted that the interior of the superconductor remains free of magnetic fields, due to the magnetic field produced by high current loops in the outer region of the superconductor.

2.4 Flux quantization

Cooper pairs within a superconductor can be described by the wave equation

$$\psi = \sqrt{n}\exp\left(i\varphi(\vec{r})\right) \tag{1}$$

With n representing the density and φ the phase of the Cooper pairs, the quantum mechanical current density, under the influence of an external vector potential $\vec{A}$, is expressed as:

$$\vec{j} = \frac{iq\hbar}{2m^*}\left(\psi^*\vec{\nabla}\psi - \psi\vec{\nabla}\psi^*\right) - \frac{nq^2}{m^*}\cdot\vec{A}\cdot\psi\psi^*$$

Given $q = -2e$, $m^* = 2m$, where e is the elementary charge and m is the electron mass, this leads to the following expression [1]:

$$\vec{j} = -\frac{\hbar n e}{m} \cdot \vec{\nabla}\varphi - \frac{2ne^2}{m} \cdot \vec{A}$$

We now examine the current density within a superconducting ring traced along a closed path:

$$\oint \vec{j} \cdot d\vec{s} = -\frac{\hbar n e}{m} \oint \vec{\nabla}\varphi \cdot d\vec{s} - \frac{ne^2}{m} \oint \vec{A} \cdot d\vec{s} \tag{2}$$

The wave function's phase must remain unchanged after a full rotation, meaning the closed curve integral must equal an integer multiple of 2π:

$$\oint \vec{\nabla}\varphi \cdot d\vec{s} = 2\pi N \quad \text{with} \quad N \in \mathbb{N}_0$$

This leads to

$$\frac{m}{ne^2} \oint \vec{j} \cdot d\vec{s} + \frac{2e}{\hbar} \oint \vec{A} \cdot d\vec{s} = -2\pi N \tag{3}$$

Based on the Meissner-Ochsenfeld effect, it is established that the magnetic field $\vec{B}$ inside the superconductor is zero. Using the Maxwell equation $\boldsymbol{\nabla} \times \vec{B} = \mu_0 \vec{j}$, it follows that $\vec{j} = 0$. The first term in equation 3 therefore vanishes. Consequently, it is derived using Stokes' theorem that:

$$\oint_{\partial F} \vec{A} \cdot d\vec{s} = \iint_F \boldsymbol{\nabla} \times \vec{A} \cdot d\vec{f} = \iint_F \vec{B} \cdot d\vec{f} = \Phi$$

Where F represents the area enclosed by the curve and Φ denotes the magnetic flux. Inserting this relation into equation 3 yields:

$$\Phi = -\frac{h}{2e} N.$$

It is evident that the magnetic flux within a superconducting ring is quantized, and the presence of $-2e$ indicates that the charge carriers consist of paired electrons [2, Chapter 10.9].

2.5 Josephson effect theory

When two superconductors are linked across a weak contact, a Josephson junction forms provided the disruption is sufficiently narrow to permit the passage of Cooper pairs and unpaired electrons. The resulting current of Cooper pairs varies with the phase difference γ between the two superconductors. By assuming a weak link between them the current across the contact is the first Fourier component of the phase-depended function $f(\gamma)$ and is described by the **first Josephson equation** as [1]

$$I = I_\text{c} \sin(\gamma) \tag{4}$$

Here, $\gamma = \varphi_2 - \varphi_1 + \frac{2e}{\hbar c} \int \vec{A} \cdot d\vec{s}$, where a B-field is defined via the vector potential $\vec{A}$ through $\vec{B} = \boldsymbol{\nabla} \times \vec{A}$, and $I_\text{c}(T = 0) \sim \Delta$ for a tunneling contact. The term Δ represents the energy gap in BCS theory and is expressed as, [6, p. 9]:

$$\Delta(T = 0) = 1.76 k_B T_\text{c}.$$

When a current $I > I_\text{c}$ is applied, non-bound electrons are also transported across, leading to a resistance R_n. The resulting electric potential between the two superconductors affects the phase difference: The wave functions for the two superconductors ψ_1 and ψ_2 are expressed as [5, Chapter 21]:

$$i\hbar \frac{d}{dt} \psi_1 = V_1 \psi_1 + k \psi_2$$

$$i\hbar\frac{d}{dt}\psi_2 = V_2\psi_2 + k\psi_1$$

Here, k is a coupling constant, and $V_i = -2e\phi_i$ represents the energy potential. By applying the wave function from equation 1 for each of the two superconducting elements, it is determined that:

$$i\hbar\left(i\frac{d\varphi_1}{dt} + \frac{1}{2n_1}\frac{dn_1}{dt}\right)\psi_1 = V_1\psi_1 + k\psi_2$$
$$i\hbar\left(i\frac{d\varphi_2}{dt} + \frac{1}{2n_2}\frac{dn_2}{dt}\right)\psi_2 = V_2\psi_2 + k\psi_1$$

After dividing by the wave functions and considering the real part, it is derived that:

$$-\hbar\frac{d\varphi_1}{dt} = V_1 + k\sqrt{\frac{n_2}{n_1}}\cos(\varphi_2 - \varphi_1)$$
$$-\hbar\frac{d\varphi_2}{dt} = V_2 + k\sqrt{\frac{n_1}{n_2}}\cos(\varphi_1 - \varphi_2).$$

For a small k, this leads to:

$$\frac{d}{dt}(\varphi_2 - \varphi_1) \approx \frac{V_2 - V_1}{-\hbar} = \frac{2e}{\hbar}U\ ,$$

by using $(V_2 - V_1) = (-2e\phi_2 - (-2e\phi_1)) = -2eU$.
This is also known as the **second Josephson equation**:

$$\frac{d\gamma}{dt} - \frac{2e}{h}U = 0. \tag{5}$$

It will be shown that the second Josephson equation is gauge invariant. In Electrodynamics the electric and magnetic fields can be expressed from potentials: $\vec{E} = -\vec{\nabla}\phi - \frac{\partial\vec{A}}{\partial t}$ and $\vec{B} = \vec{\nabla}\times\vec{A}$. The electric and magnetic fields stay the same under these gauge transformations: The electric potential ϕ transforms as

$$\phi \to \phi - \frac{1}{c}\frac{d\chi}{dt}$$

where χ is the gauge function. The transformation of $\vec{A}$ is

$$\vec{A} \to \vec{A} + \vec{\nabla}\chi$$

If the second Josephson equation is gauge invariant φ must transform as

$$\varphi \to \varphi - \frac{2e}{\hbar c}\chi.$$

Because $U = \phi_2 - \phi_1$ and $\gamma = \varphi_2 - \varphi_1$, we see that under a gauge transformation of the second Josephson equation:

$$\frac{\mathrm{d}\gamma}{\mathrm{d}t} - \frac{2e}{\hbar}U = 0$$

$$\Leftrightarrow \frac{\mathrm{d}}{\mathrm{d}t}\left(\varphi_2 - \varphi_1 - \frac{2e}{\hbar c}\chi_2 + \frac{2e}{\hbar c}\chi_1\right) - \frac{2e}{\hbar}\left(\phi_2 - \phi_1 - \frac{1}{c}\frac{\mathrm{d}\chi_2}{\mathrm{d}t} + \frac{1}{c}\frac{\mathrm{d}\chi_1}{\mathrm{d}t}\right) = 0$$

$$\Leftrightarrow \frac{\mathrm{d}}{\mathrm{d}t}(\varphi_2 - \varphi_1) - \frac{2e}{\hbar}(\phi_2 - \phi_1) = 0.$$

$$\Leftrightarrow \frac{\mathrm{d}\gamma}{\mathrm{d}t} - \frac{2e}{\hbar}\cdot U = 0.$$

2.6 Josephson contacts

In Figure 1 schematics for Josephson contacts are shown, highlighting their different geometries, materials and length scales. The top left corner a) shows an oxide slab between two superconductors, whereas the top right corner b) shows a slab of metal as link between the superconductors. In the two middle images in the middle c) and d) more complicated geometries are used. For c) the junction is constricted, for d) it is a double layer contact. In the bottom left corner e) a point contact between a tip and a slab of superconducting material forms the Josephson junction [1]. The current tunnels through the respective contact. There is

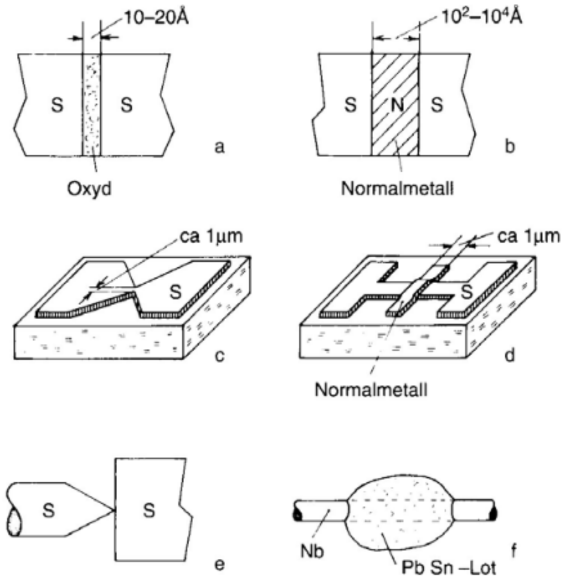


Figure 1: Different possible scenarios for the construction of Josephson junctions.[1]

a correlation between the tunneling current I and the magnetic flux Φ in Josephson contacts because the phase difference γ is influenced by the vector potential $\vec{A}$. This correlation is described by the equation

$$I_c(H) = I_c(0) \cdot \left| \frac{\sin(\pi\Phi/\Phi_0)}{\pi\Phi/\Phi_0} \right| ,$$

where $\Phi = 2\Lambda_l l B$, Φ_0 is the flux quantum and l is the size of the contact perpendicular to the magnetic field [1].

2.7 DC - Josephson effect

In the case of no voltage drop across the Josephson junction, $\gamma =$ const. in time as can be seen from the second Josephson equation 5. This implies for the current across the junction that $I = I_c \sin\gamma =$ const. if $|I| < I_c$ [1]. This is a DC current across the junction permitted by tunneling.

2.8 AC - Josephson effect

If there is a voltage drop U across the Josephson contact the phase difference $\gamma = \gamma_0 + (2e/\hbar)Ut$ from equation 5. It then follows from equation 4 that the current across the junction is:

$$I = I_c \cdot \sin\left(\gamma_0 + \frac{2e}{\hbar}Ut\right) \tag{6}$$

This is an AC current with alternating frequency $\omega = 2\pi f = 2e/\hbar \cdot U$. This can be interpreted as the tunneling Cooper pairs being accelerated by the voltage and emitting this excess energy as radiation of energy $E = \hbar\omega = 2eU$ [1]. If the emitted radiation has a frequency of $f = 10\,\text{GHz}$, the voltage drop is

$U = (hf)/(2e) = \frac{1}{483.6\,\mathrm{MHz}/\mu\mathrm{V}} \cdot 10\,\mathrm{GHz} = 20.68\,\mu\mathrm{V}$. The second harmonic of this emission lies at a frequency of $f_{2\mathrm{nd}} = 2 \cdot (2e/h \cdot U_{2\mathrm{nd}}) = 10\,\mathrm{GHz}$. This corresponds to a voltage of $U_{2\mathrm{nd}} = 1/2 \cdot U = 10.34\,\mu\mathrm{V}$.

2.9 Current-voltage characteristics of Josephson contacts

To model the current across the Josephson contact, also incorporating normally conducting electrons and the capacitance between the electrodes, the RSJ model is used. Figure 2 shows the circuit diagram which contains the superconducting current, the resistance due to the normally conducting electrons R_n and the capacitance C. The total current I_tot flowing through the Josephson contact is then:

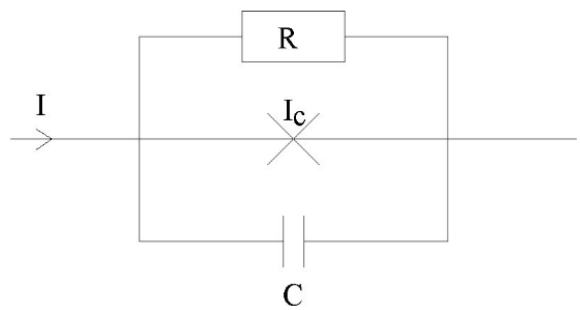


Figure 2: An equivalent circuit diagram for the RSJ-model [1]

$$I_\mathrm{tot} = I_\mathrm{c}\sin\gamma + \frac{U}{R_\mathrm{n}} + C\frac{\mathrm{d}U}{\mathrm{d}t} \tag{7}$$

The terms in I_tot can be rewritten by using the second Josephson equation 5, such that $U = \hbar/(2e)\frac{\mathrm{d}\gamma}{\mathrm{d}t}$ and $\frac{\mathrm{d}U}{\mathrm{d}t} = \hbar/(2e)\frac{\mathrm{d}^2\gamma}{\mathrm{d}t^2}$. This leads to an equation for the total current in terms of γ:

$$I_\mathrm{tot} = I_\mathrm{c}\sin\gamma + \frac{\hbar}{2eR_\mathrm{n}}\frac{\mathrm{d}\gamma}{\mathrm{d}t} + \frac{\hbar C}{2e}\frac{\mathrm{d}^2\gamma}{\mathrm{d}t^2} \tag{8}$$

For small phase differences $\sin\gamma \approx \gamma$ and equation 8 looks similar to the equation for a driven damped harmonic oscillator $F(t) = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 2\lambda\frac{\mathrm{d}x}{\mathrm{d}t} + \omega_0^2 x$, where the damping parameter $\lambda = \frac{1}{2R_\mathrm{n}C}$, the eigenfrequency $\omega_0^2 = \frac{2eI_\mathrm{c}}{\hbar C}$ and $F(t) = \frac{2eI_\mathrm{tot}}{\hbar C}$. It An important parameter which describes the evolution of the current-voltage curve is the McCumber parameter

$$\beta_\mathrm{c} = \frac{2\pi C R_\mathrm{n}^2 I_\mathrm{c}}{\Phi_0}\ , \tag{9}$$

where $\Phi_0 = 2\pi\hbar/e$ is the flux quantum. This parameter determines whether the current-voltage curve is under- or overdamped.
$\beta_\mathrm{c} > 1$:
Here the system is underdamped and displays hysteresis. If one starts at an external current I_tot below I_c no voltage is measured across the Josephson junction. In this case the Cooper pairs cross the Josephson contact by tunneling. Once I_c is reached, there is a voltage appearing. Now the Cooper pairs fall apart and the behavior of the I-U curve is like a resistor. If one decreases the external current again, the voltage stays even if $I_\mathrm{tot} < I_\mathrm{c}$ until a certain retrap current I_R is reached. This hysteresis can be used to determine the McCumber-parameter:

$$\frac{I_\mathrm{R}}{I_\mathrm{c}} = \frac{4}{\pi\sqrt{\beta_\mathrm{c}}} \tag{10}$$

$\beta_\mathrm{c} < 1$:
In this case the system is overdamped and does not display hysteresis. If the capacitance is zero the differential equation can be solved analytically. In this "weak link" case the Cooper pairs already fall apart at lower currents. [1]

2.10 4-point resistance measurement

To precisely measure the resistance the 4-point measurement method needs to be used. The advantage of this method is that the resistance of the cables and thermal noise from them can be eliminated from the measurement which is necessary at low temperatures. Figure 3 shows a circuit with a 4-point resistance measurement setup. It works by sending a current through the device which is supposed to be measured from point 1 to point 4. The voltage drop across points 2 and 3 is measured. Due to the high internal resistance of the voltmeter almost no current flows through it. The measured resistance is then $R = U_{23}/I_{14}$.

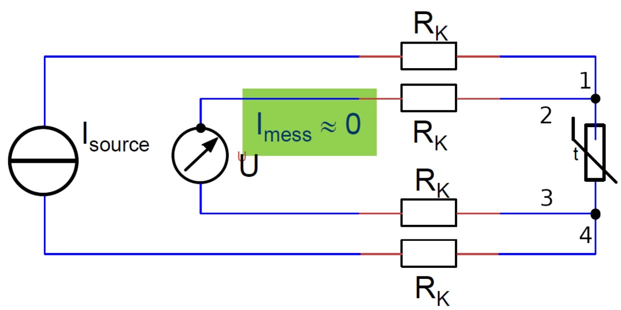


Figure 3: 4-point method resistance measurement circuit [4]

2.11 Experimental setup

The Josephson point contact which is used in this experiment is placed on a dipstick along with measurement equipment, see Figure 4. A horn antenna measures the radiation coming from the Josephson contact and its signal is further transported through a waveguide which is effective in the frequency range 9.8 GHz − 11.5 GHz to a Radiometer. The dipstick can be lowered into a tank of liquid helium (critical temperature ≈ 4 K) for cooling as the superconductor niobium has a critical temperature of ≈ 9.3 K. The distance between the niobium tip and the niobium slab can be adjusted as well.

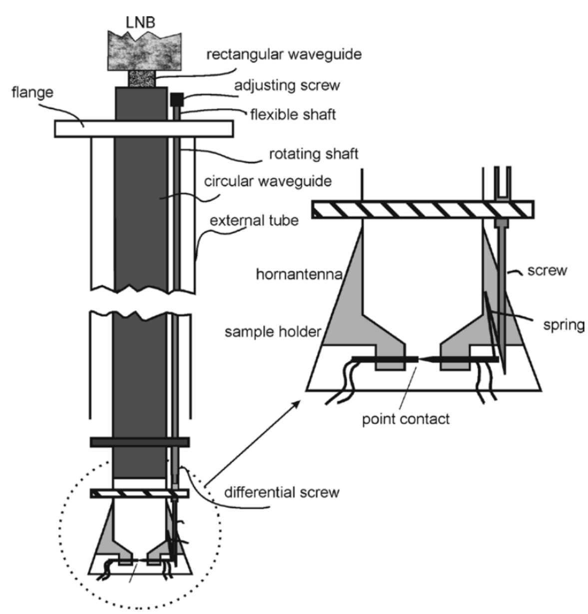


Figure 4: Schematic of the experimental setup [1]

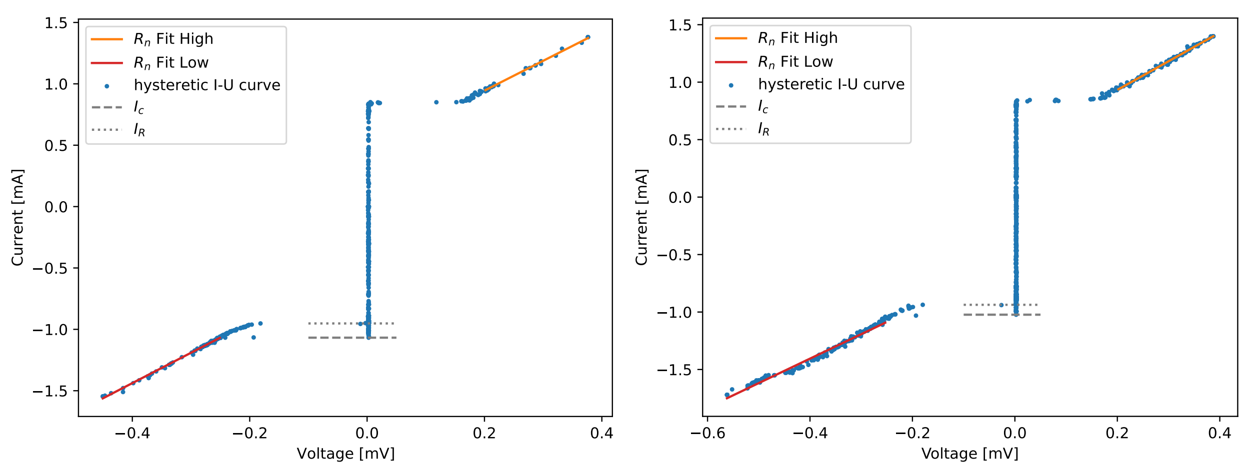
3 Experimental Procedure

In order to use the niobium tip and the niobium slab as a Josephson junction, the tip has to be prepared first. For this the tip is removed from the dipstick and sharpened by using a piece of sandpaper. After sharpening and cleaning the tip is placed back into the dipstick. The distance between tip and slab is calibrated by measuring the current through the junction such that this current is high enough. Now the dipstick gets placed and fixed into the liquid helium dewar. To investigate the superconductive behaviour of the Josephson junction the dipstick is slowly lowered into the helium dewar to cool down to niobiums critical temperature of 9.3 K. The temperature is monitored by a PT100-temperature sensor, which gives reliable measurements down to $\sim 40\,\mathrm{K}$. Below this point the voltage-current characteristic is monitored to reach the superconductive state. There are voltage-current measurements taken for two different states. One at a colder temperature, where hysteretic behaviour is observed and one at a slightly higher temperature without hysteresis. For the state without hysteresis a radiometer measures the electromagnetic emission from the Josephson junction. The measurement of hysteretic behaviour was successful, whereas older data was used for the other state. A possible reason for the failure to measure the emission from the non-hysteretic state could be a not-sharp enough or dirty niobium tip.

4 Analysis

4.1 Hysteretic current-voltage characteristic

In Figure 5 we plot the current-voltage characteristic at a temperature where the behaviour of the Josephson junction is hysteretic. To determine the current in Ampere across the junction the measured voltage is multiplied by the voltage scale, here 1 mA. As the external current across the junction is increased the measured voltage drop remains zero as long as the current is below the critical current I_c as the Cooper-pairs tunnel through the Josephson junction. When I_c is surpassed normal conducting electrons also travel across the junction, leading to a normal resistance R_n and voltage drop U. When the temperature is low enough, the system is underdamped and hysteresis occurs.



(a) First hysteretic measurement

(b) Second hysteretic measurement

Figure 5: Hysteretic current-voltage characteristic curves in blue. Two linear fits at higher and lower voltage to determine the normal resistance above the critical current. Indicated as well are the critical current T_c and the retrap current I_R.

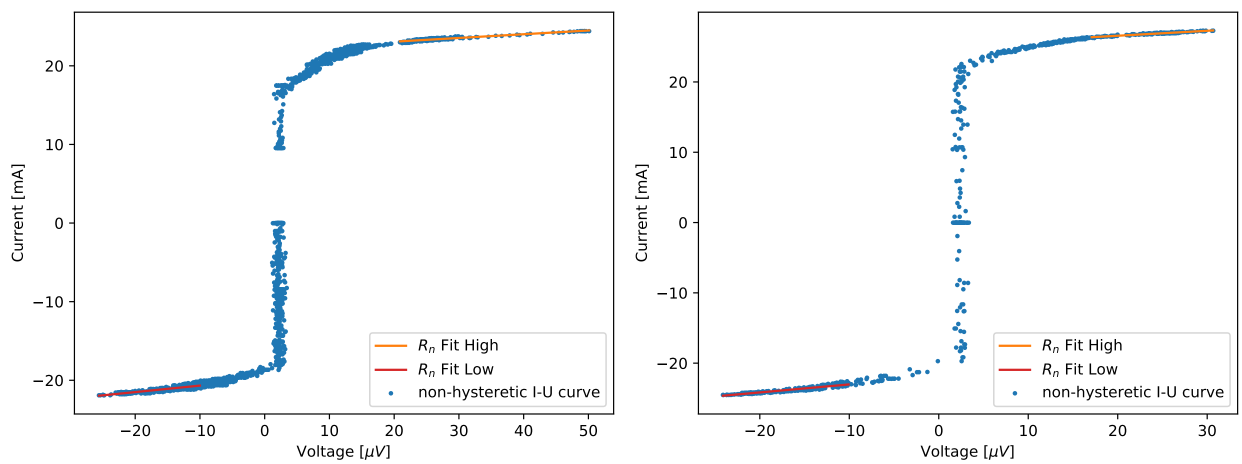
The normal resistance is determined by fitting a linear function $I(U) = a \cdot U + b$ to the higher and lower end of the voltage. This leads to the values:

$$\begin{aligned}
&\text{Measurement 1}:\\
&\qquad \mathrm{U} < 0: R_\mathrm{n} = 404.5 \pm 2.3\,\mathrm{m}\Omega\\
&\qquad \mathrm{U} > 0: R_\mathrm{n} = 412 \pm 4\,\mathrm{m}\Omega\\
&\text{Measurement 2}:\\
&\qquad \mathrm{U} < 0: R_\mathrm{n} = 470 \pm 4\,\mathrm{m}\Omega\\
&\qquad \mathrm{U} > 0: R_\mathrm{n} = 401.2 \pm 1.8\,\mathrm{m}\Omega\ ,
\end{aligned}$$

where the errors are determined from the fit.

4.2 Non-hysteretic current-voltage characteristic

The data for the non-hysteretic curve was taken from a previous group. Here the current conversion scale was 100 mA. Figure 6 shows two of their measured curves. Here no hysteretic behaviour is observed. This is because the measurement is taken at a slightly higher temperature than the hysteretic curve. This higher temperature leads to less resistance for the normal-conducting electrons, due to them having more thermal energy, which leads to an overdamped system and to no hysteresis.



(a) First non-hysteretic measurement

(b) Second non-hysteretic measurement

Figure 6: Non-hysteretic current-voltage characteristic curve in blue. There are two linear fits at higher and lower voltages to determine the normal resistance above the critical current.

Here the normal resistance R_n is determined in analogy to the hysteretic curve. This leads to the values:

$$\begin{aligned}
&\text{Measurement 1}:\\
&\qquad \mathrm{U} < 0: R_\mathrm{n} = 21.1 \pm 0.5\,\mathrm{m}\Omega\\
&\qquad \mathrm{U} > 0: R_\mathrm{n} = 12.1 \pm 0.4\,\mathrm{m}\Omega\\
&\text{Measurement 2}:\\
&\qquad \mathrm{U} < 0: R_\mathrm{n} = 8.78 \pm 0.22\,\mathrm{m}\Omega\\
&\qquad \mathrm{U} > 0: R_\mathrm{n} = 14.17 \pm 0.25\,\mathrm{m}\Omega
\end{aligned}$$

This resistance is lower than the resistance for the hysteretic case but a comparison is not feasible as the data comes from a measurement from a previous group.

4.3 McCumber parameter and self-capacitance

The McCumber parameter β_c is a descriptive parameter of the RCSJ-model, which can distinguish between the non-hysteretic and hysteretic case. It can be determined from the two hysteretic curves in Figure 5 and the equation:

$$\beta = \left(\frac{4I_\mathrm{c}}{\pi I_\mathrm{R}}\right)^2 \ . \tag{11}$$

First we need to determine the critical current I_c and the retrap current I_R. For this the two $U < 0$ parts are used from the measurements in Figure 5 as the $U > 0$ parts do not really show hysteretic behaviour. Those are the highest (absolute) current in the current increasing phase and the lowest (absolute) current in the current decreasing phase as shown in Figure 5. The values for I_c and I_R are:

$$\begin{aligned}
\text{Plot 1}: I_\mathrm{c} &= -1.068\,\mathrm{mA} \\
I_\mathrm{R} &= -0.952\,\mathrm{mA} \\
\text{Plot 2}: I_\mathrm{c} &= -1.023\,\mathrm{mA} \\
I_\mathrm{R} &= -0.937\,\mathrm{mA} \\
\text{Average}: \langle I_\mathrm{c}\rangle &= -1.045 \pm 0.023\,\mathrm{mA} \\
\langle I_\mathrm{R}\rangle &= -0.944 \pm 0.008\,\mathrm{mA}
\end{aligned}$$

Finally the McCumber parameter β_c is:

$$\beta_\mathrm{c} = 1.99 \pm 0.10 \ , \tag{12}$$

where the error is calculated according to Gaussian error propagation:

$$\Delta\beta_\mathrm{c} = \sqrt{\left(\left(\frac{4}{\pi I_\mathrm{R}}\right)^2 \cdot 2I_C \cdot \Delta I_\mathrm{c}\right)^2 + \left(\left(\frac{4I_\mathrm{c}}{\pi}\right)^2 \cdot \frac{-2}{I_\mathrm{R}^3} \cdot \Delta I_\mathrm{R}\right)^2} \tag{13}$$

In the hysteretic state the system is underdamped, which leads to $\beta_\mathrm{c} > 1$. This is measured here to be true. What can be determined as well is the self-capacitance C from the equation

$$C = \frac{\beta_\mathrm{c}\Phi_0}{2\pi R_\mathrm{n}^2 I_\mathrm{c}} \tag{14}$$

With $R_\mathrm{n}\rangle = 422.0 \pm 2.9\,\mathrm{m\Omega}$ from averaging the four values from Figure 5, $I_\mathrm{c} = 1.045 \pm 0.023\,\mathrm{mA}$ and $\Phi_0 = h/(2e)$, this leads to:

$$C = 3.51 \pm 0.19\,\mathrm{pF} \ , \tag{15}$$

where the error comes from Gaussian error propagation:

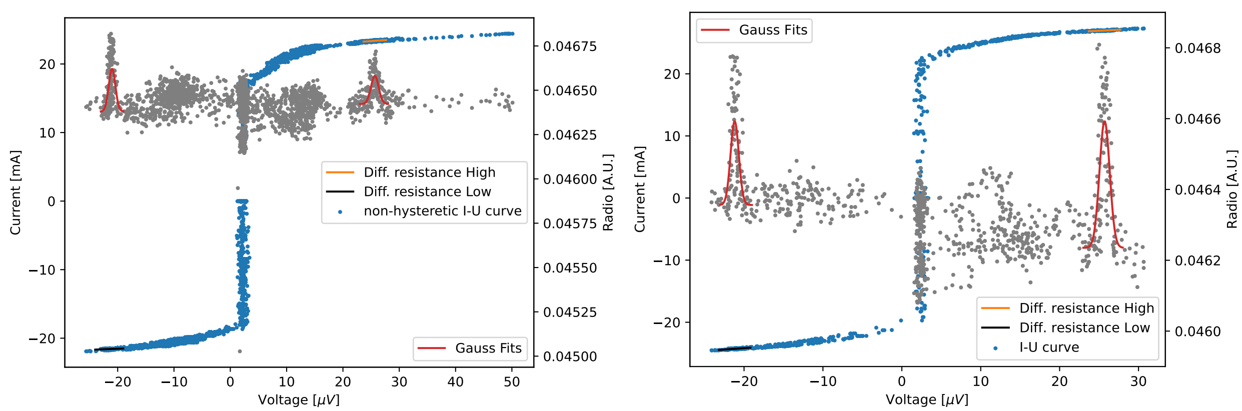
$$\Delta C = \sqrt{\left(\frac{\Phi_0}{2\pi I_\mathrm{c} R_\mathrm{n}^2} \cdot \Delta\beta_\mathrm{c}\right)^2 + \left(\frac{-\beta_\mathrm{c}\Phi_0}{\pi I_\mathrm{c} R_\mathrm{n}^3} \cdot \Delta R_\mathrm{n}\right)^2 + \left(\frac{-\beta_\mathrm{c}\Phi_0}{2\pi I_\mathrm{c}^2 R_\mathrm{n}^2} \cdot \Delta I_\mathrm{c}\right)^2} \tag{16}$$

4.4 Analysis of emission maxima

In Figure 7 there are the non-hysteretic voltage-current curve as well as simultaneous radiometer measurements shown. At a particular voltage there is an emission maximum detected in the small range where the radiometer is sensitive. The emission is due to Cooper pairs emitting the excess energy gained by the voltage across the Josephson junction, once I_c is exceeded. First we determine the differential resistance $\mathrm{d}U/\mathrm{d}I$ at the emission maximum. The voltage at the maximum is determined from a Gaussian fit, as described in subsection 4.5. For $\mathrm{d}U/\mathrm{d}I$ a linear function $I(U, a, b) = aU + b$ is fitted to the current-voltage characteristic around the voltages, where the emission maximum lies. $\mathrm{d}U/\mathrm{d}I$ is then the inverse of the parameter a of the fit-function. This can be seen in Figure 7. It follows that:

$$
\begin{aligned}
&\text{Measurement 1}: \\
&\qquad \mathrm{U} < 0: \frac{\mathrm{d}U}{\mathrm{d}I} = 32 \pm 8\,\mathrm{m}\Omega \\
&\qquad \mathrm{U} > 0: \frac{\mathrm{d}U}{\mathrm{d}I} = 17.5 \pm 2.3\,\mathrm{m}\Omega \\
&\text{Measurement 2}: \\
&\qquad \mathrm{U} < 0: \frac{\mathrm{d}U}{\mathrm{d}I} = 11.2 \pm 1.0\,\mathrm{m}\Omega \\
&\qquad \mathrm{U} > 0: \frac{\mathrm{d}U}{\mathrm{d}I} = 15.9 \pm 1.3\,\mathrm{m}\Omega\,.
\end{aligned}
$$

It can be seen from measurement 1 that a higher differential resistance leads to a narrower emission line width. This is visible in Figure 7 a) when the negative voltage and positive voltage emission peaks are compared as well as their c-fit parameters from the Gaussian fit as determined in subsection 4.5, where c^2 is smaller for the $U < 0$ case than the $U > 0$ case. This is in contradiction to the results from the second measurement, where a smaller differential resistance leads to a narrower fit, if their respective c^2 and $\mathrm{d}U/\mathrm{d}I$ values are compared. Considering that the uncertainties on measurement 2 values are smaller, this scenario is more likely to be true. The result from measurement 1 could maybe be explained by the Gaussian fit not being suitable for the data in that emission peak due to a less dense coverage toward the lower voltages of the $U < 0$ fit. Expected is that a lower differential resistance leads to a lower amount of damping and therefore a narrower emission peak. This is the case for measurement 2.



(a) First non-hysteretic measurement

(b) Second non-hysteretic measurement

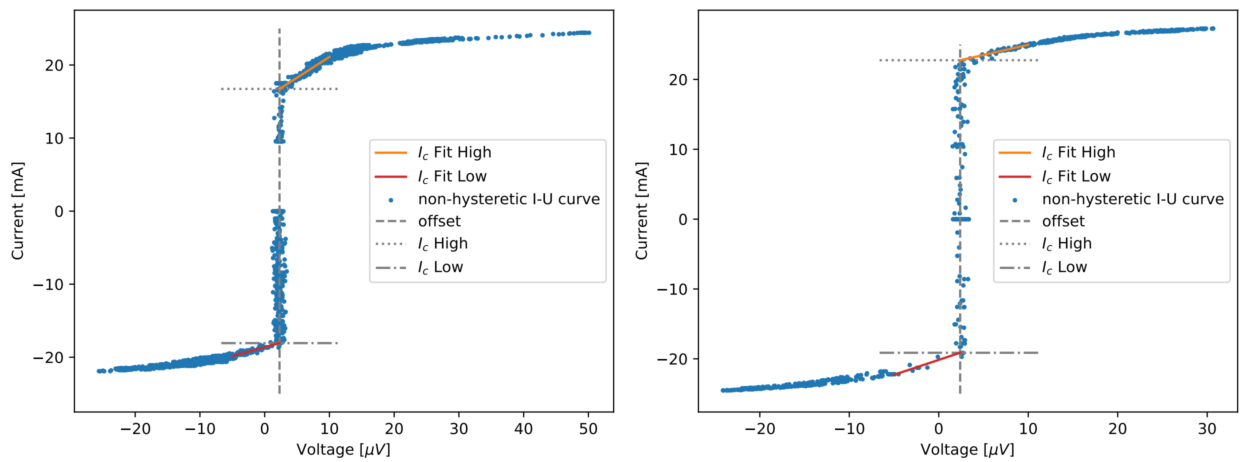
Figure 7: Non-hysteretic current-voltage curve in blue with radiometer measurements in grey. Gaussian fits to maximum emission range to indicate the maxium emission voltage in red as well as linear fits to the current-voltage curve in orange and black to determine the differential resistance at the maximum emission voltage.

Next we determine the critical current I_c for the measurements in Figure 8. To do this we fit a linear function $I(U, a, b) = aU + b$ to the high- and low current-voltage curve and evaluate the fitted function at the zero-voltage (minus offset) point. The offsets are $U_{\text{Offset}} = 2.29\,\mu\text{V}$ for measurement 1 and $U_{\text{Offset}} = 2.39\,\mu\text{V}$

for measurement 2 as determined in subsection 4.5. This results in the values:

$$\begin{aligned}
&\text{Measurement 1}: \\
&\qquad \mathrm{U} < 0 : I_\mathrm{c} = -18.1\,\mathrm{mA} \\
&\qquad \mathrm{U} > 0 : I_\mathrm{c} = 16.7\,\mathrm{mA} \\
&\text{Measurement 2}: \\
&\qquad \mathrm{U} < 0 : I_\mathrm{c} = -19.1\,\mathrm{mA} \\
&\qquad \mathrm{U} > 0 : I_\mathrm{c} = 22.7\,\mathrm{mA} \\
&\text{Average}: \\
&\qquad \langle I_\mathrm{c} \rangle = 19.2 \pm 0.6\,\mathrm{mA}\ ,
\end{aligned}$$

Comparing this critical current to the one from the hysteretic curve is not useful, as the non-hysteretic curve comes from an previous group which did not use the exact same niobium-tip setup.



(a) First non-hysteretic measurement

(b) Second non-hysteretic measurement

Figure 8: Non-hysteretic current-voltage curve with two linear fits to determine I_c in red and orange. The positions of the zero-point offset U_offset, the positive bias critical current $I_{\mathrm{c,High}}$ and the negative bias critical current $I_{\mathrm{c,Low}}$ are also indicated.

4.5 Determination of $2e/h$

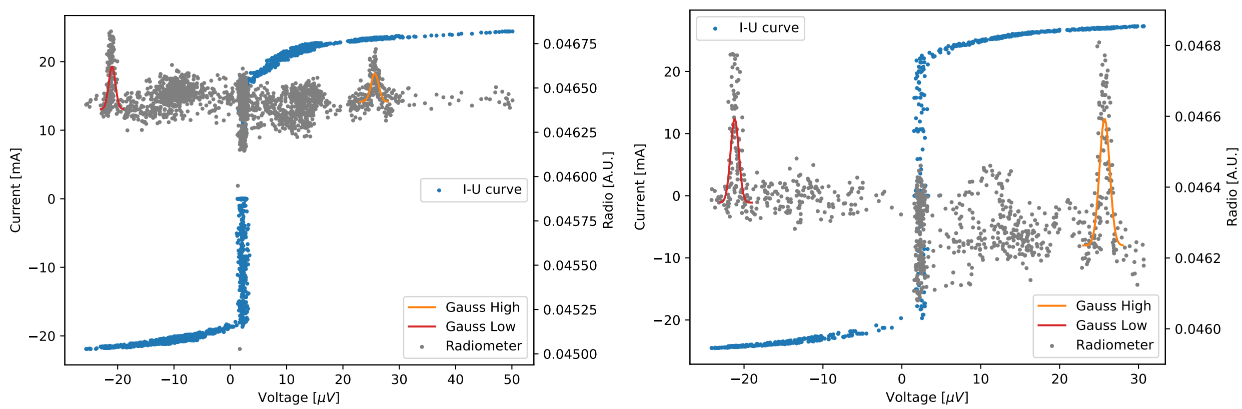
The value of $2e/h$ can be determined by analysing the emission coming from the Josephson junction in the non-hysteretic state. $2e/h$ can be calculated with the equation

$$\frac{2e}{h} = \frac{f}{U}\ , \tag{17}$$

where f is the emission frequency and U is the voltage. Figure 9 shows the non-hysteretic curve with radiometer measurements. The frequency of this measurement was $f = 11.3\,\mathrm{GHz}$ and U is determined by fitting the Gaussian function

$$f(U, a, b, c, d) = a \exp\left(-\frac{1}{2}\frac{(U-b)^2}{c^2}\right) + d\ , \tag{18}$$

to the emission maxima.



(a) First non-hysteretic measurement

(b) Second non-hysteretic measurement

Figure 9: Non-hysteretic current-voltage curve in blue with additionally the radiometer measurements in grey. In red and orange are Gaussian fits to the emission maxima in the radio measurements to determine the maxium emission voltage.

An offset of $U_{\text{Offset}} = 2.29\,\mu\text{V}$ and $U_{\text{Offset}} = 2.39\,\mu\text{V}$ for measurement 1 and 2 respectively is applied, determined by averaging the voltage around the purely superconducting state of the current-voltage curve. This results in the voltage values from the fit-parameter b and the fit parameter c^2 as an indicator for the emission peak width:

$$\begin{aligned}
&\text{Measurement 1}: \\
&\qquad \text{U} < 0: U = -23.29 \pm 0.06\,\mu\text{V} && c^2 = 351 \pm 11\,\text{fV} \\
&\qquad \text{U} > 0: U = 23.36 \pm 0.06\,\mu\text{V} && c^2 = 462 \pm 9\,\text{fV} \\
&\text{Measurement 2}: \\
&\qquad \text{U} < 0: U = -23.58 \pm 0.06\,\mu\text{V} && c^2 = 274 \pm 8\,\text{fV} \\
&\qquad \text{U} > 0: U = 23.31 \pm 0.05\,\mu\text{V} && c^2 = 418 \pm 6\,\text{fV} \\
&\quad \text{Average}: \\
&\qquad\quad \langle U \rangle = 23.38 \pm 0.06\,\mu\text{V} && ,
\end{aligned}$$

where the errors come from the fit. This finally leads to:

$$\frac{2e}{h} = \frac{f}{U} = 483.2 \pm 1.2\,\text{MHz}/\mu\text{V}\ , \tag{19}$$

where the error comes from error propagation:

$$\Delta(2e/h) = \frac{f}{U^2} \cdot \Delta U\ . \tag{20}$$

The literature value of $2e/h = 483.6\,\text{MHz}/\mu\text{V}$ lies within a 1-σ range of this measurement. The measurement is therefore in agreement.

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